WHY C IS THE MOTHER OF ALL PROGRAMMING LANGUAGES

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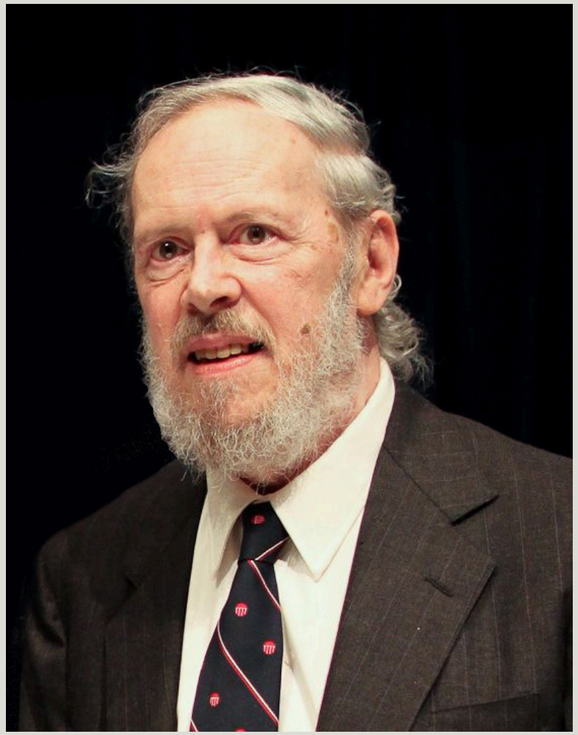

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INTRODUCTION TO C

- C is a general-purpose programming language
- Developed in 1972 at Bell Laboratories
- Created by Dennis Ritchie
- Designed for system programming and operating systems
- Known for speed, efficiency, and simplicity

1960s: Precursors (1)

ALGOL, BCPL, and B languages lay the groundwork for structured programming concepts.

1972: Birth of C (2)

Dennis Ritchie develops C at Bell Labs, inheriting features from B and BCPL.

1973: Unix Rewrite (3)

The Unix kernel is largely rewritten in C, proving its system programming capabilities.

1978: K&R C (4)

Brian Kernighan and Dennis Ritchie publish "The C Programming Language," establishing a de facto standard.

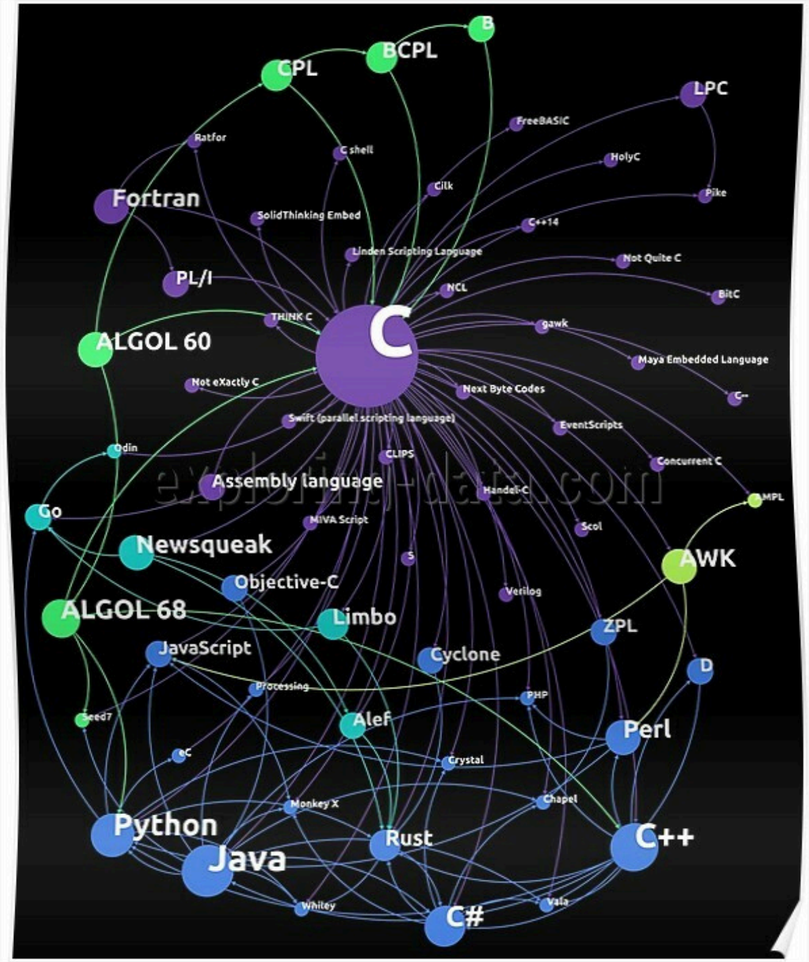
1989: ANSI C Standard (5)

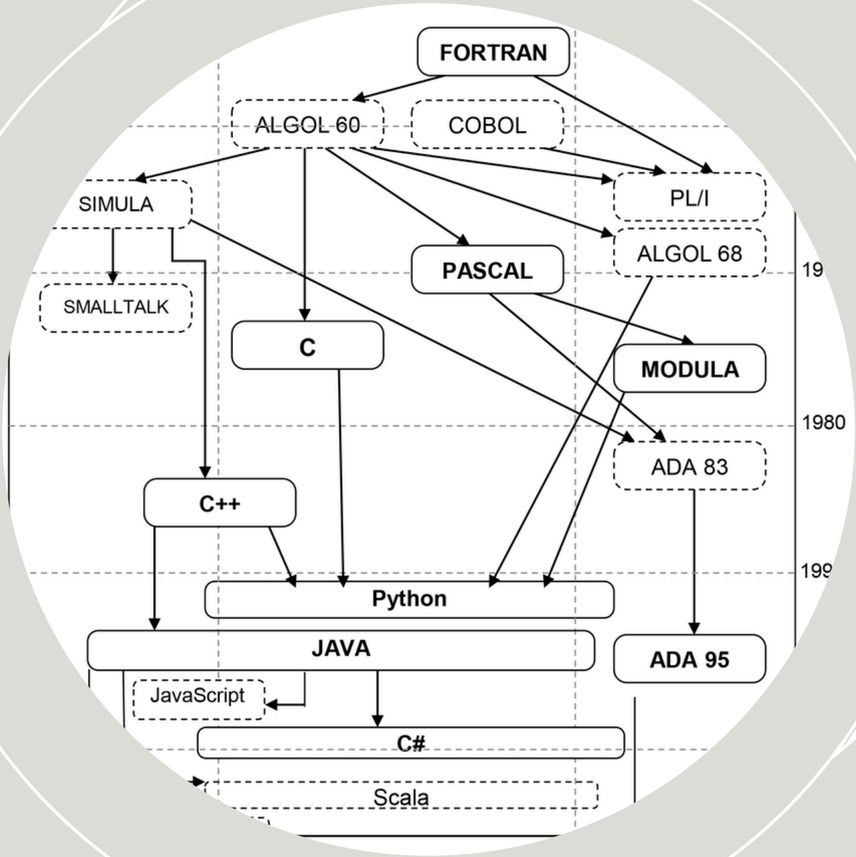
C is formally standardized by the American National Standards Institute (ANSI), ensuring portability.

WHY IS C CALLED THE "MOTHER LANGUAGE"?

C is called the mother of all programming languages because:

- Many modern languages are directly or indirectly based on C
- C introduced core concepts still used today
- It laid the foundation of system-level programming





LANGUAGES DERIVED FROM C

Examples of languages influenced by C:

- C++ – Extension of C with OOP features
- Java – Syntax inspired by C
- Python – Uses C-based runtime
- C#, Objective-C, Go, Rust

C++

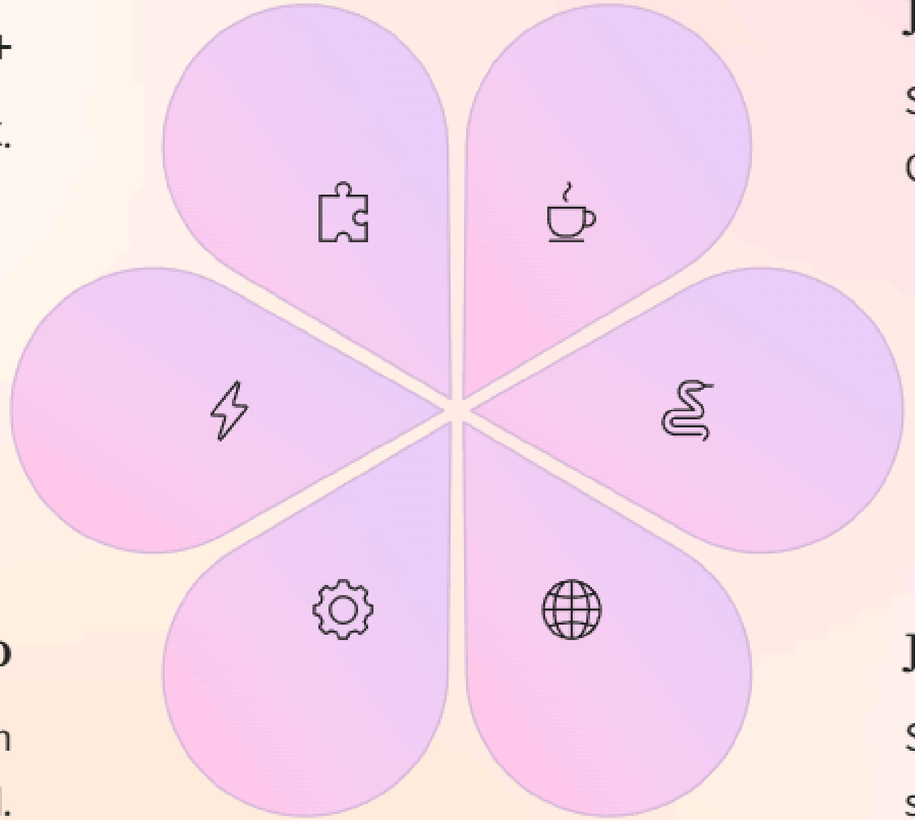
Object-oriented extensions built directly on C.

Rust

Aims for C's performance with modern memory safety.

Go

Designed with C's efficiency and simplicity in mind.



Java

Syntax and basic structure heavily influenced by C/C++.

Python

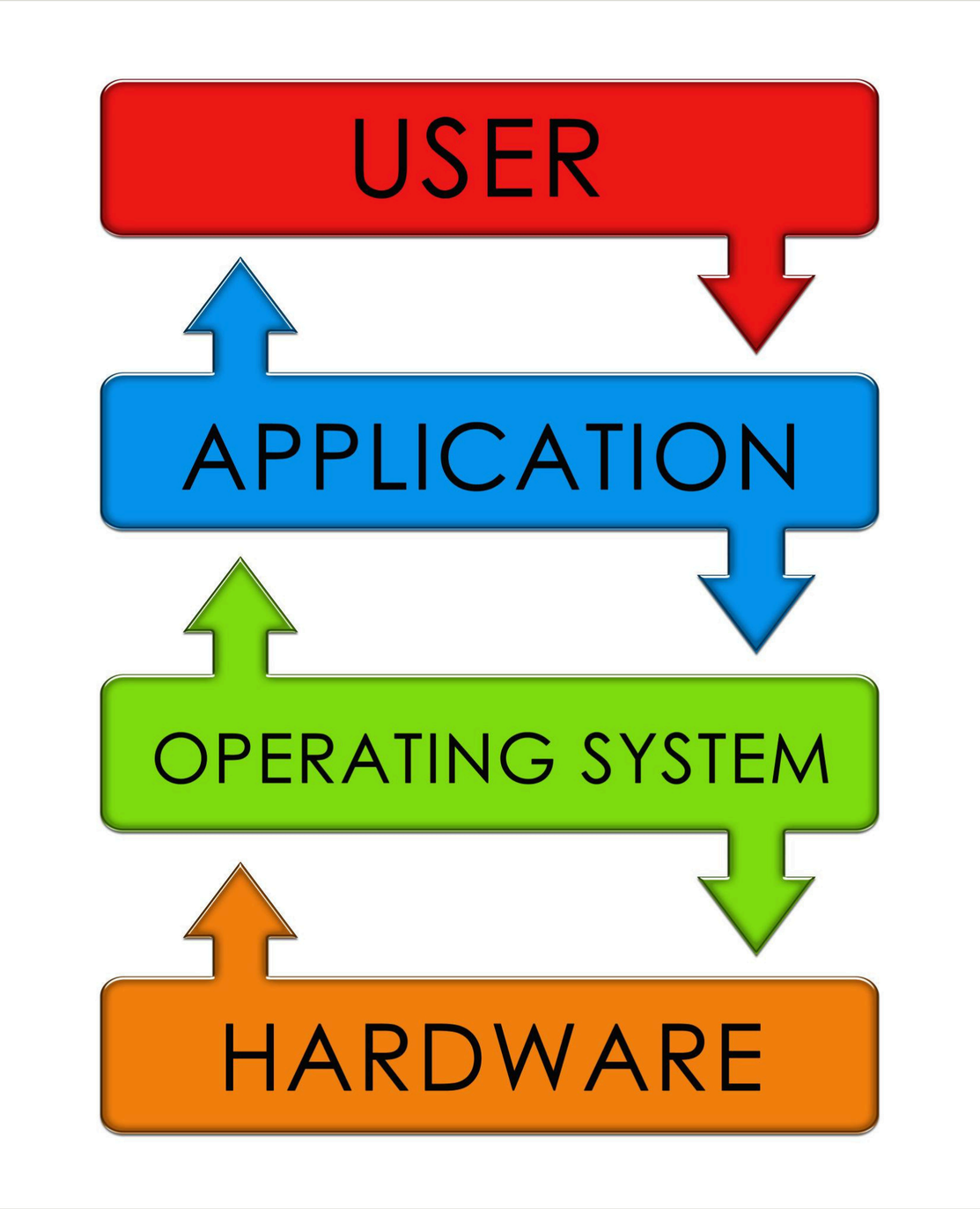
Many interpreters and core libraries written in C.

JavaScript

Shares C-like syntax, especially for control structures.

FOUNDATION OF OPERATING SYSTEMS

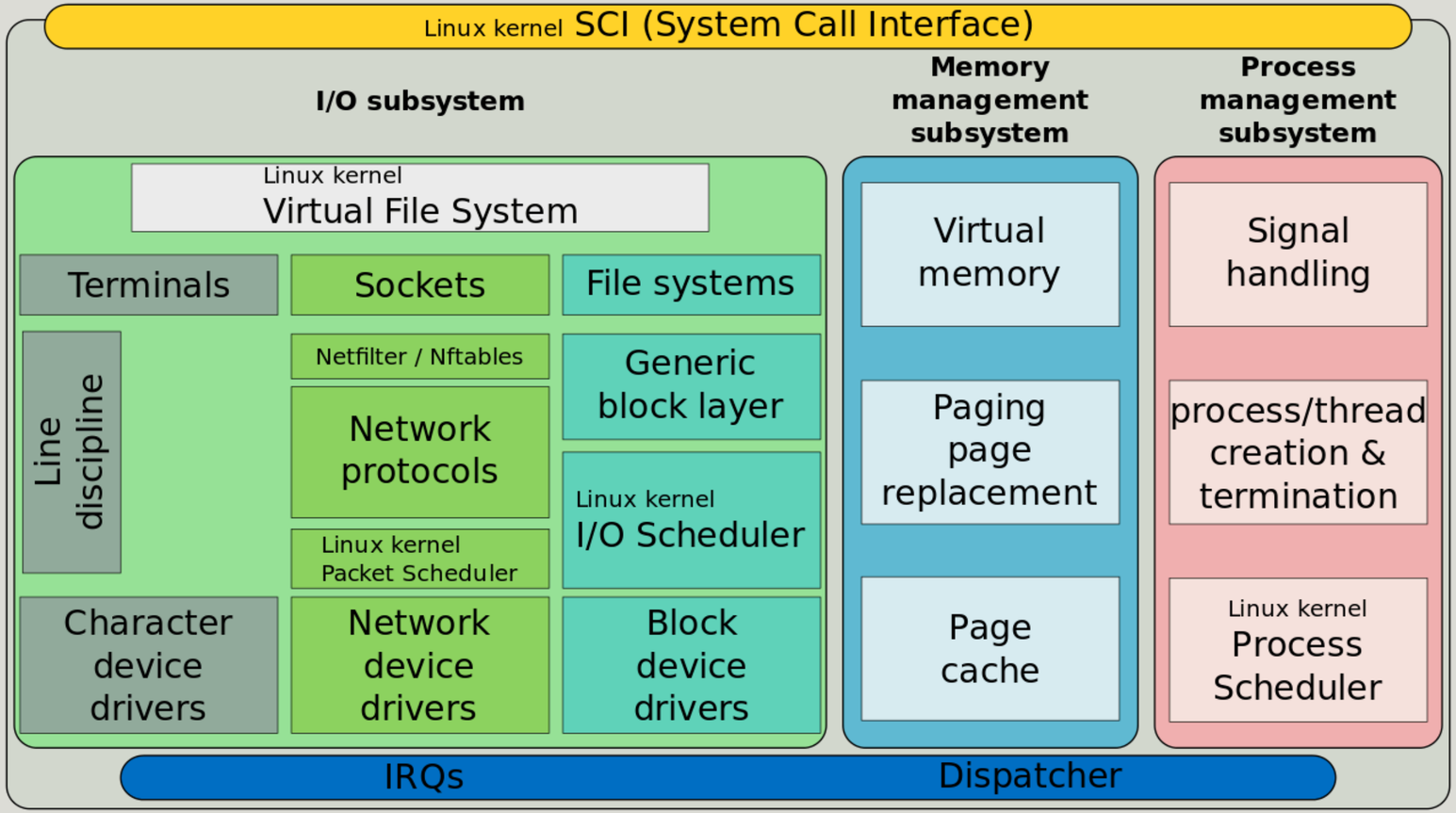
- UNIX was written in C
- Linux Kernel is mostly written in C
- Windows, macOS, Android use C internally
- C provides direct hardware access

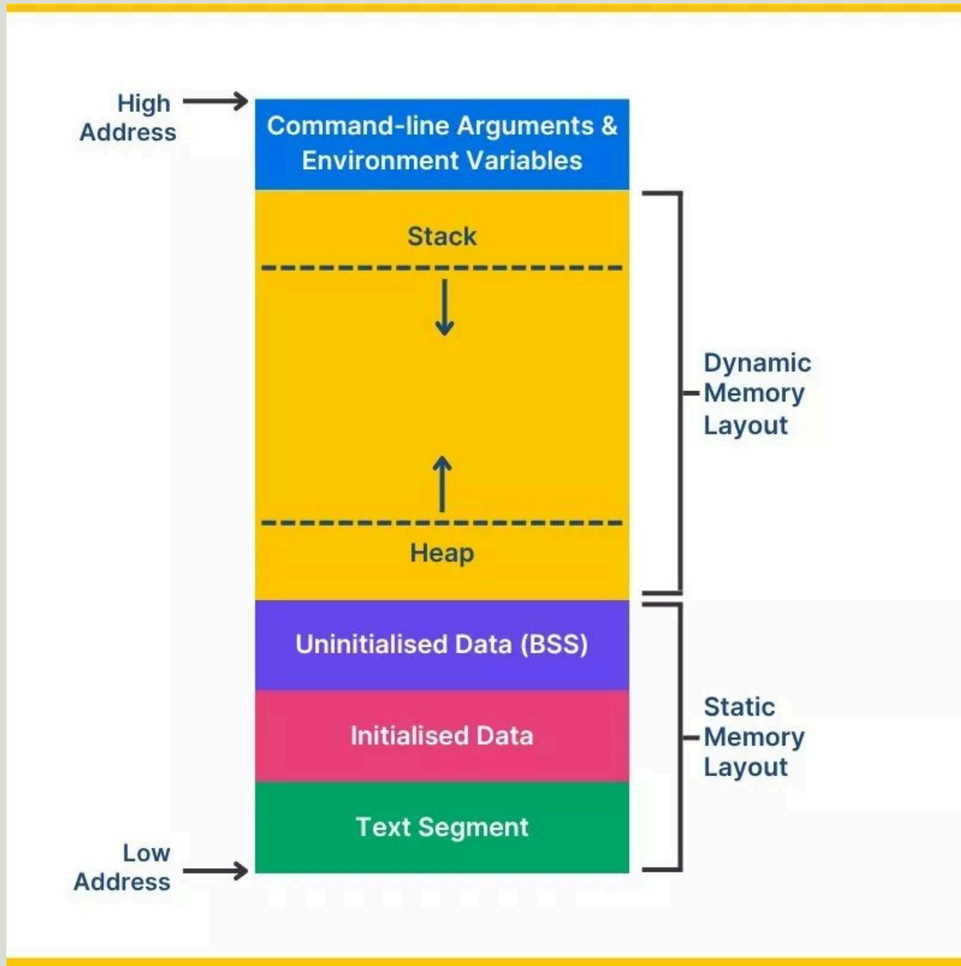


SPEED, EFFICIENCY, AND CONTROL

- Fast execution speed
- Manual memory management
- Close to hardware level
- Minimal runtime overhead

➡️ Ideal for embedded systems, drivers, and kernels



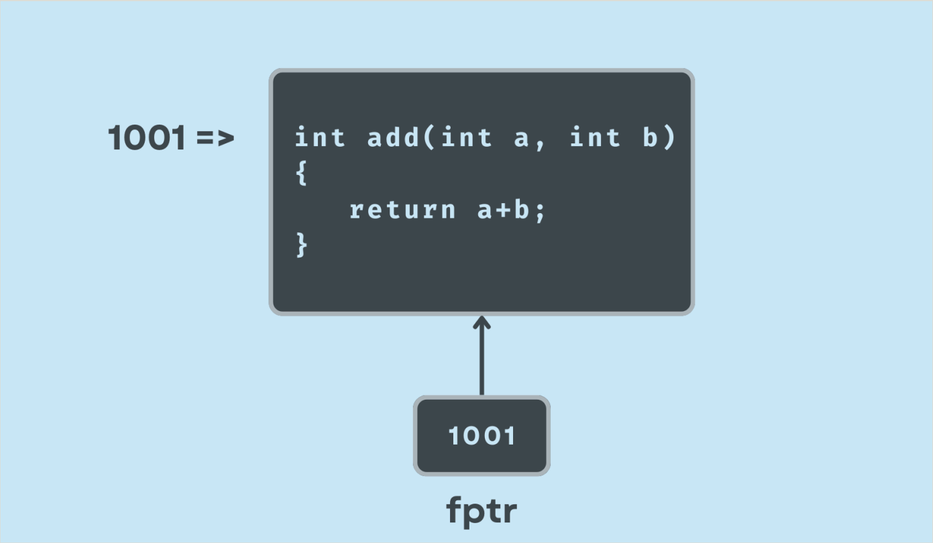


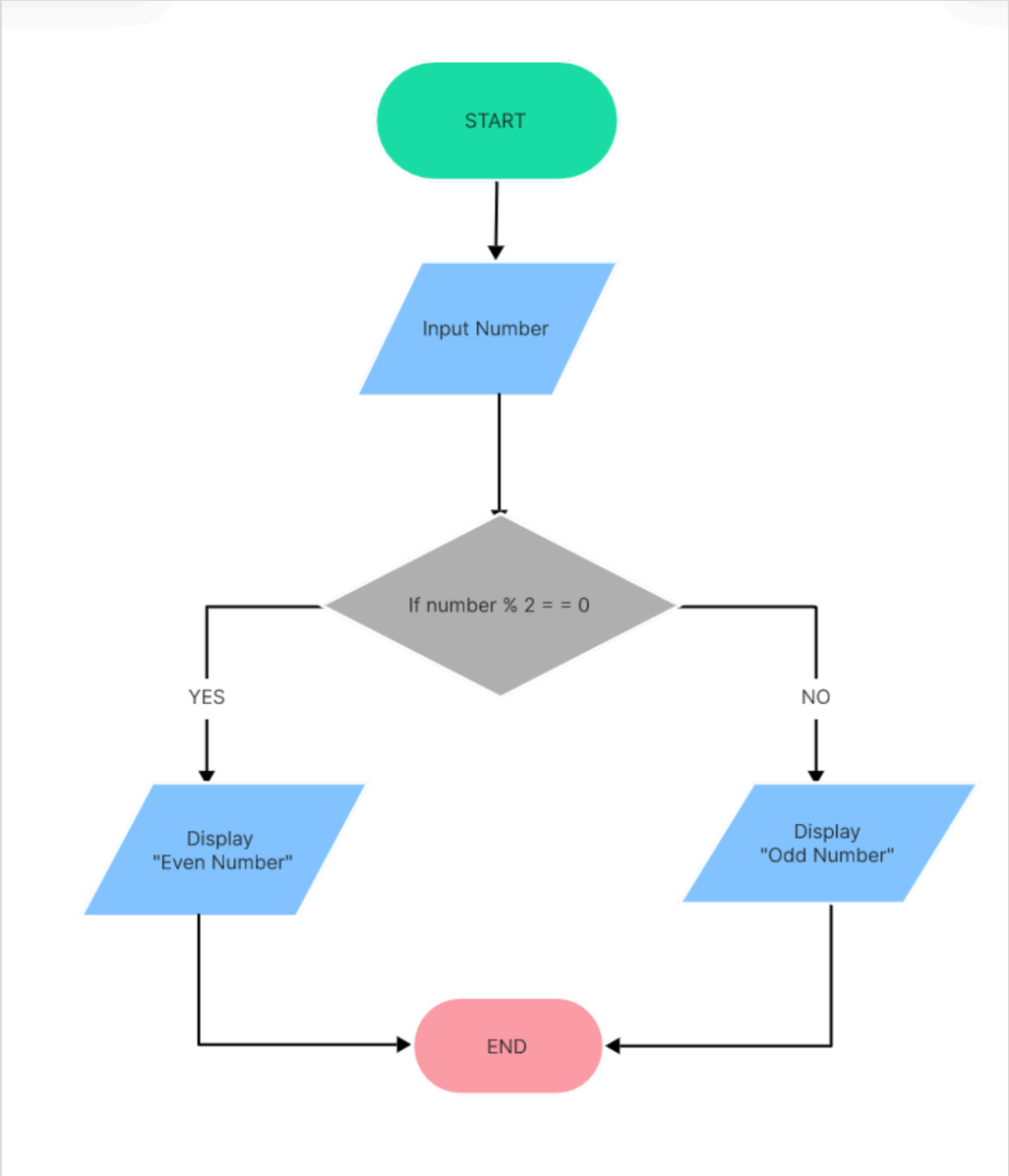
CORE CONCEPTS INTRODUCED BY C

C introduced important concepts:

- Functions
- Pointers
- Arrays
- Structures
- Bitwise operations

➡️ These concepts exist in almost every modern language



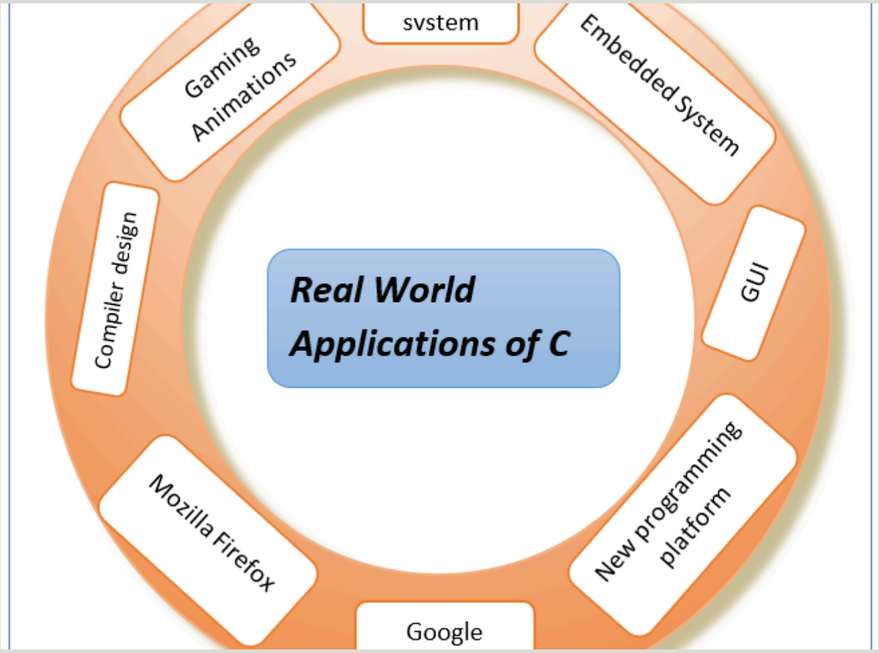


USE OF C IN REAL WORLD

C is used in:

- Embedded systems
- Microcontrollers
- Compilers & interpreters
- Game engines
- Networking software

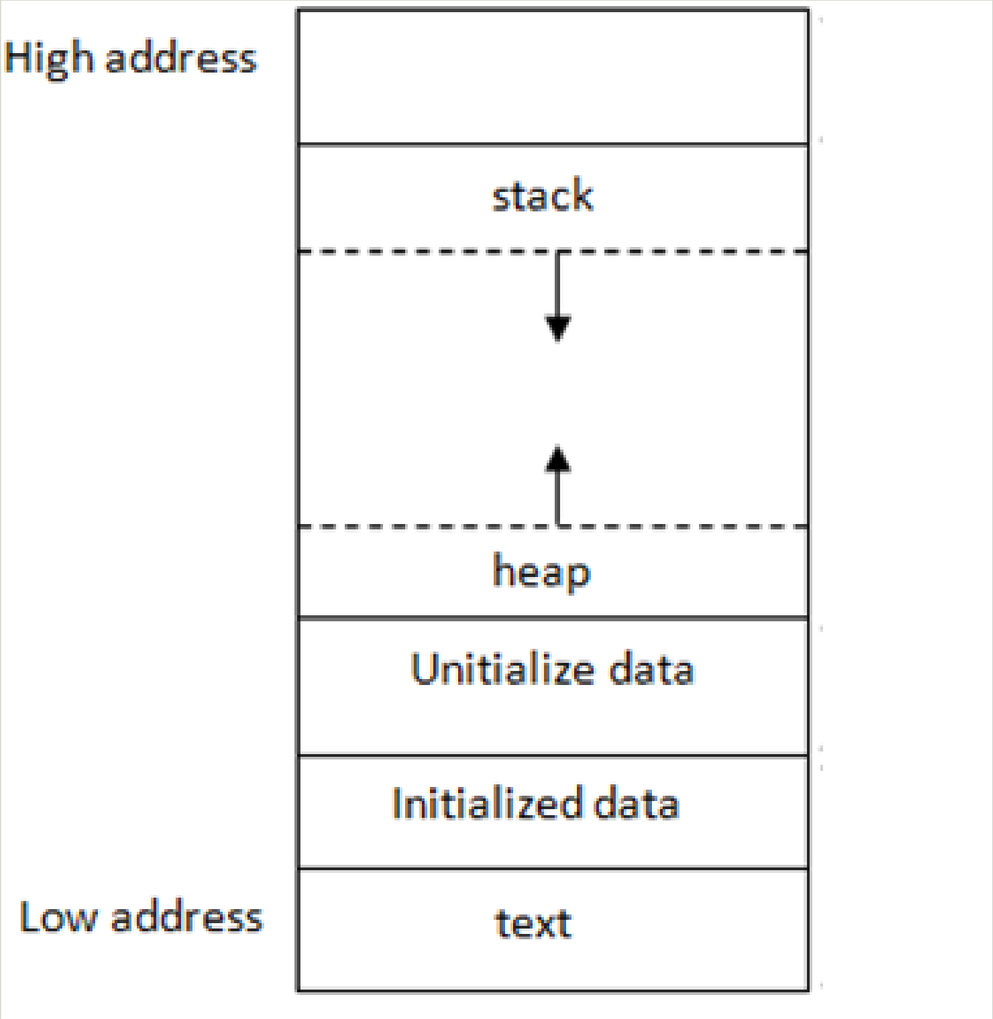
➡️ Still widely used after 50+ years



WHY C IS STILL TAUGHT FIRST

- Builds strong programming fundamentals
- Improves logical thinking
- Helps understand how computers work
- Acts as a base for learning other languages
- C forces programmers to manually manage memory providing invaluable insight into how computers stores and retrives data.

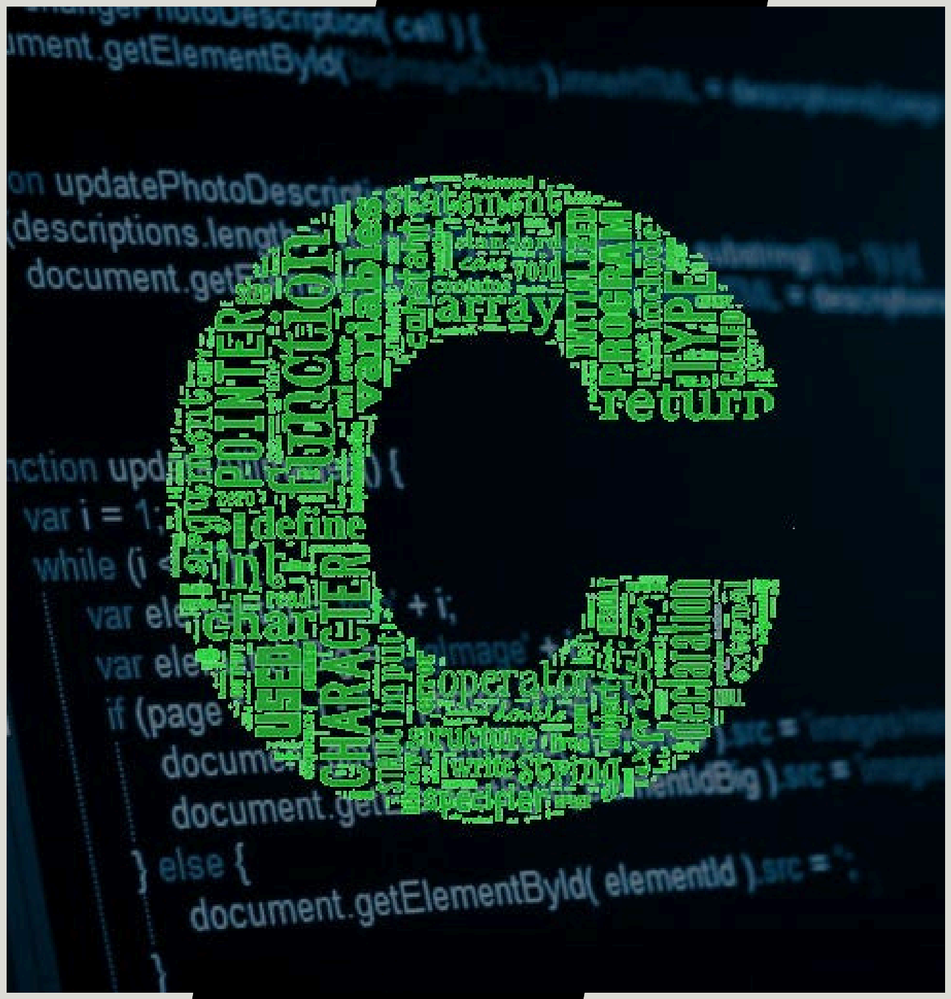
➡️ “If you know C, you can learn any language.”



CONCLUSION

- C shaped the modern programming world
- Influenced almost every major language
- Powers operating systems and hardware
- Timeless, powerful, and foundational

THEREFORE. C IS RIGHTLY CALLED THE "MOTHER OF ALL PROGRAMMING LANGUAGES."



Fortran
Cobol
LISP
BASIC
Algol 60
PL/1
Simula
ML
C
Algol 68
Scheme
Pascal
Smalltalk
Dylan
Ada
C++
Perl
Modula 3
Prolog
Java
C#
Python
Ruby

THANK YOU

Contributions

1. Pranab Kumar Singh - Origin and history of C.
2. Naman Verma - Core features of C
3. Shashwat Shivesh - Foundation for other languages
4. Leeshan Thakre - System level and real world usage
5. Naimish Akhare - Educational Value & Conclusion